

Lecture 5: CPU Scheduling

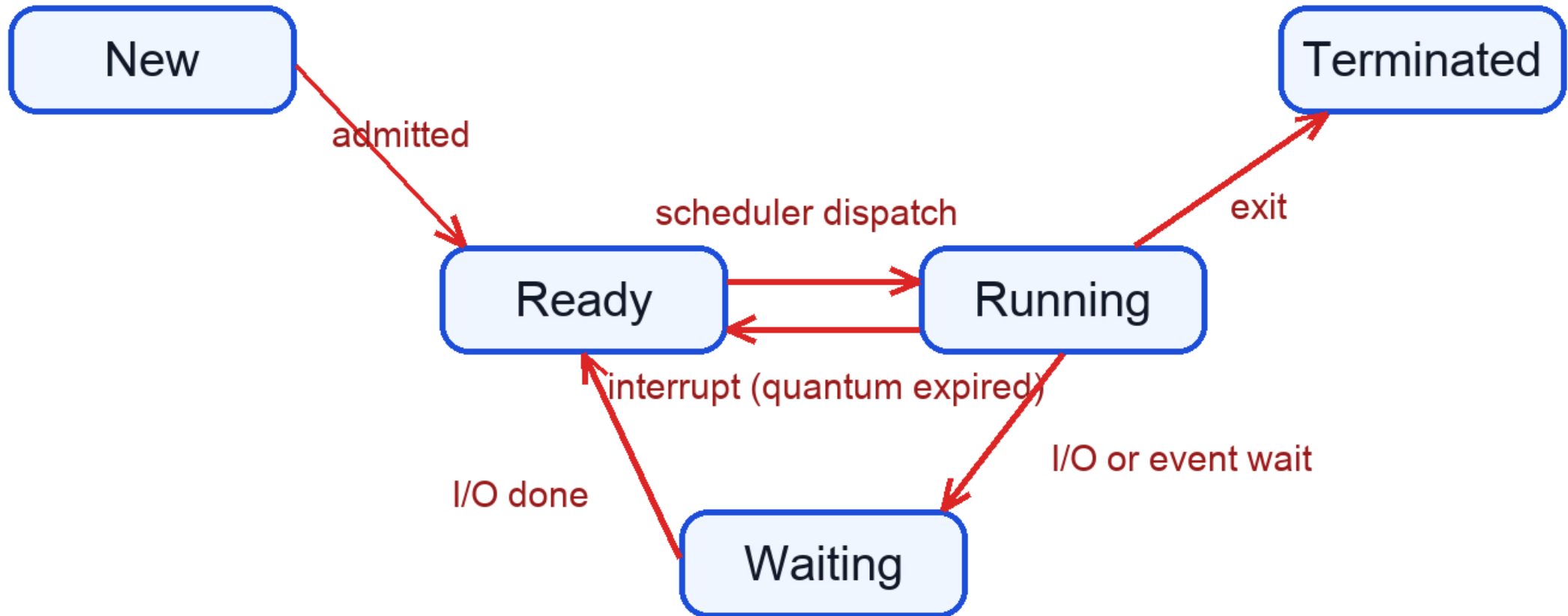
OS 101 · Operating Systems

Agenda

- Process states and the scheduler
- Scheduling criteria
- FCFS, SJF, Round Robin
- Predicting the next CPU burst
- Summary and practice questions

Process States

- The scheduler moves processes between states (see diagram)



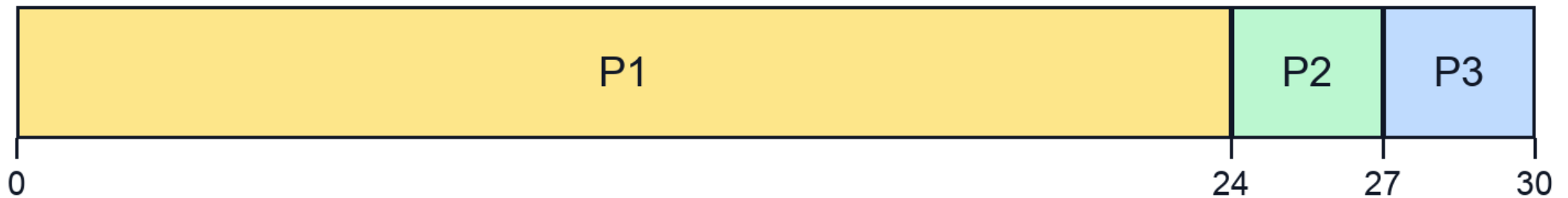
Scheduling Criteria

- CPU utilization: keep the CPU as busy as possible
- Throughput: # of processes completed per time unit
- Turnaround time = completion time - arrival time
- Waiting time = turnaround time - burst time
- Response time: time until the first response is produced

First-Come, First-Served (FCFS)

- Bursts: P1 = 24, P2 = 3, P3 = 3
- Average waiting time = $(0 + 24 + 27) / 3 = 17$
- Convoy effect: short processes wait behind a long one

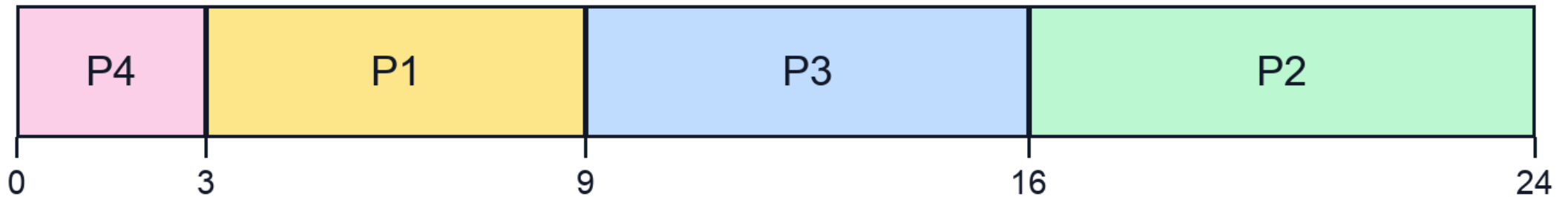
FCFS order: P1, P2, P3 (all arrive at $t=0$)



Shortest-Job-First (SJF)

- Pick the process with the smallest next CPU burst
- Provably optimal for average waiting time
- Problem: we cannot know the next burst length in advance

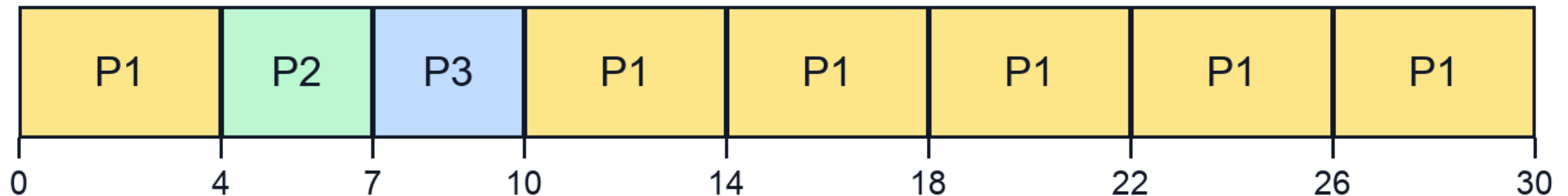
SJF (non-preemptive), all arrive at $t=0$



Round Robin (RR)

- Each process gets a small unit of CPU time (time quantum)
- After the quantum expires, the process is preempted to the ready queue
- Read the quantum off the Gantt chart below

Round Robin, time quantum $q = ?$



Predicting the Next CPU Burst

- Exponential averaging:
 - $\tau(n+1) = \alpha * t(n) + (1 - \alpha) * \tau(n)$
 - $t(n)$: actual length of the n -th CPU burst
 - $\tau(n+1)$: predicted value for the next CPU burst
 - $0 \leq \alpha \leq 1$ (commonly $\alpha = 1/2$)

Summary & Practice

- FCFS is simple but suffers from the convoy effect
- SJF minimizes average waiting time but needs burst prediction
- RR is fair; performance depends heavily on the quantum size
- Q1. Compute the average waiting time for the RR example (slide 7)
- Q2. With $\alpha = 1/2$, $\tau(0) = 10$, $t(0) = 6$: what is $\tau(1)$?